Script-Controlled ACL – Restrict Record Access Based on Field Value Project

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Project Title: Script-Controlled ACL – Restrict Record Access Based on Field Value

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Institution: Madurai Sivakasi Nadars Pioneer Meenakshi Women's College

District: Sivagangai

College code: Aluau13

Department/Year: Information Technology(BCA)– 3rd Year

Branch: BCA

Semester: 5

Script-Controlled ACL – Restrict Record Access Based on Field Value

Project Overview:

A Script-Controlled ACL restricts access to records by using a custom script. The script checks specific field values and user conditions before granting access. Users can access the record only when the ACL script evaluates to true.

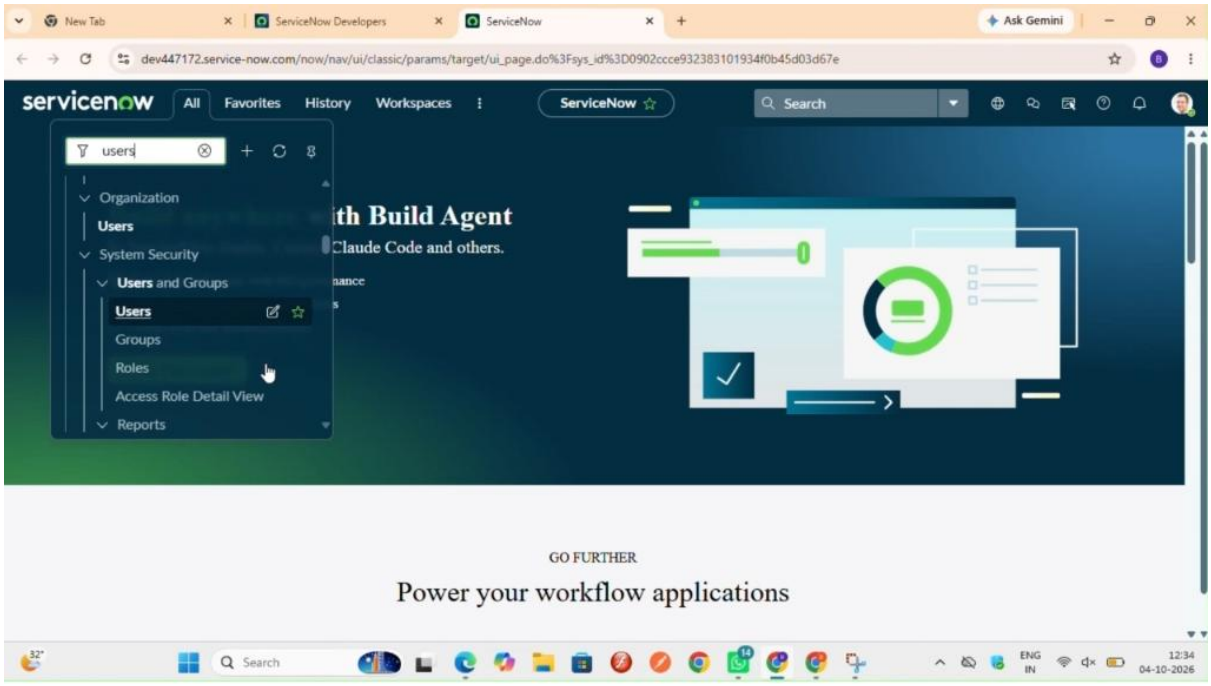
Milestone 1: Creation of Users and Roles

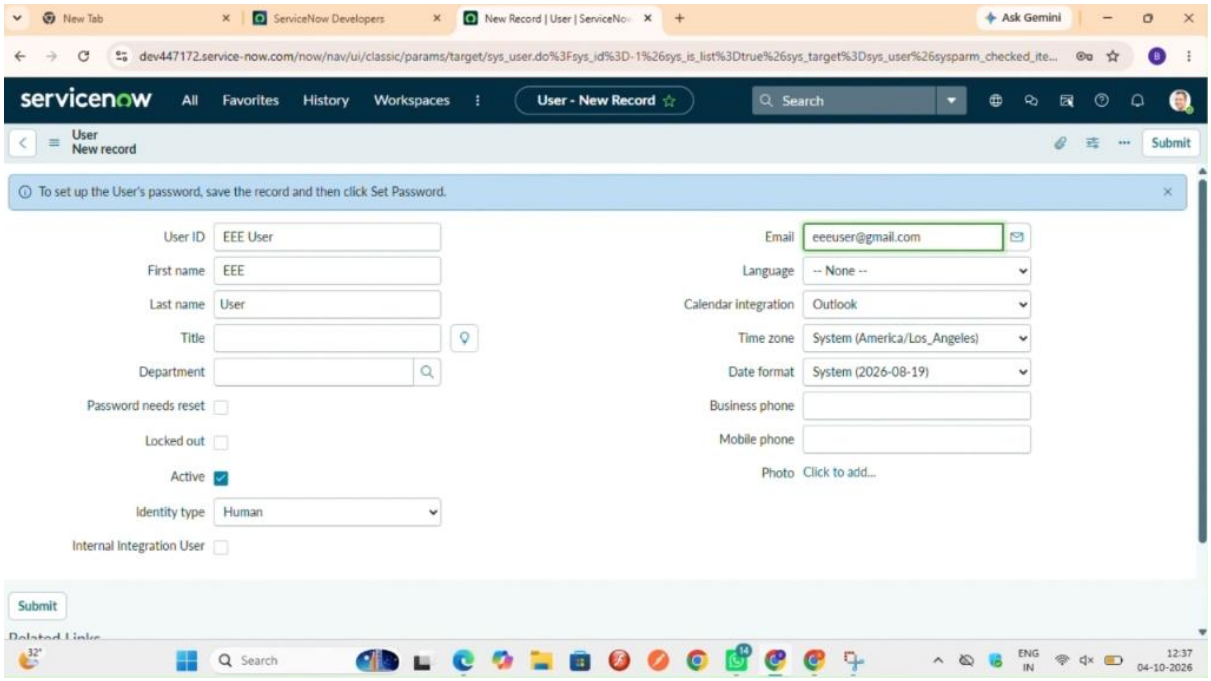
Users Creation

1. Log in to the ServiceNow instance with admin access.

2. Create a test user:

   - Navigate to User Administration → Users
   - Click New
   - User ID: EEE User
   - First name: EEE
   - Last name: User
   - Email: eeeuser@gmail.com
   - Save the record

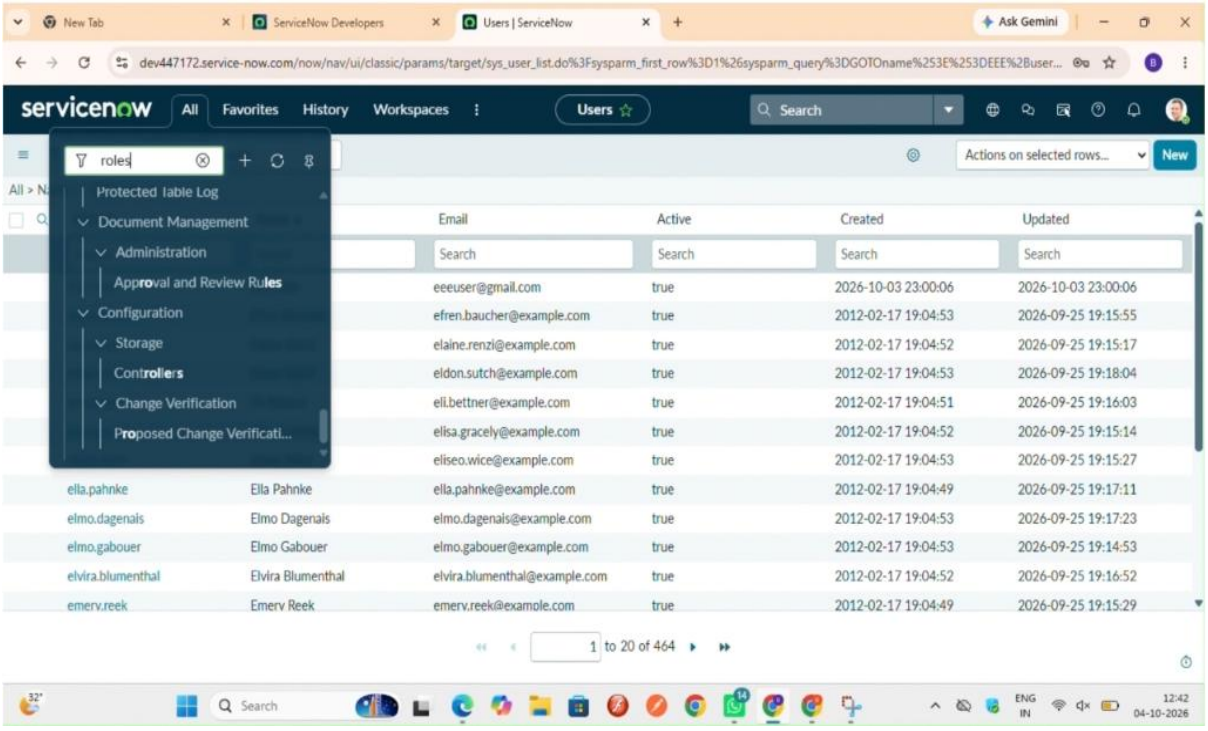


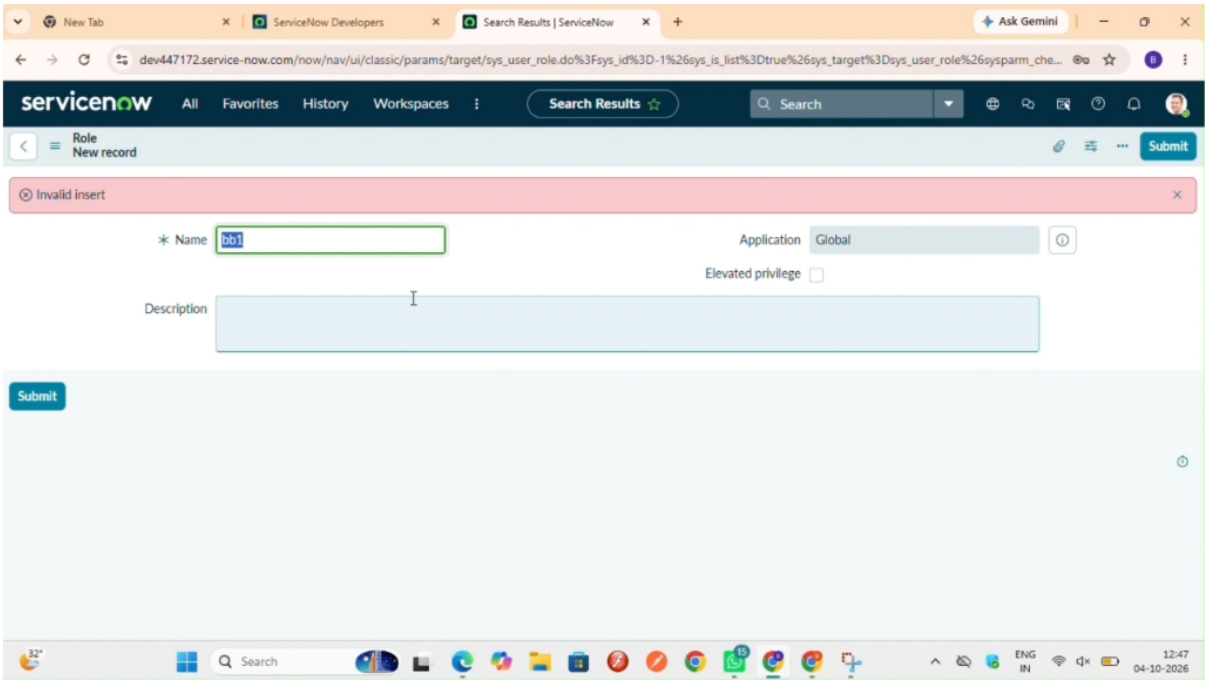


Roles Creation

- Navigate to User Administration → Roles
- Click New
- Name: Enter role name as bb1
- Description: Brief role purpose

- Click Submit
- Create other roles also like bb2,bb3,bb4.(follow the same procedure)
- Assign all the roles to the EEE User





Milestone 2: Tables Creation

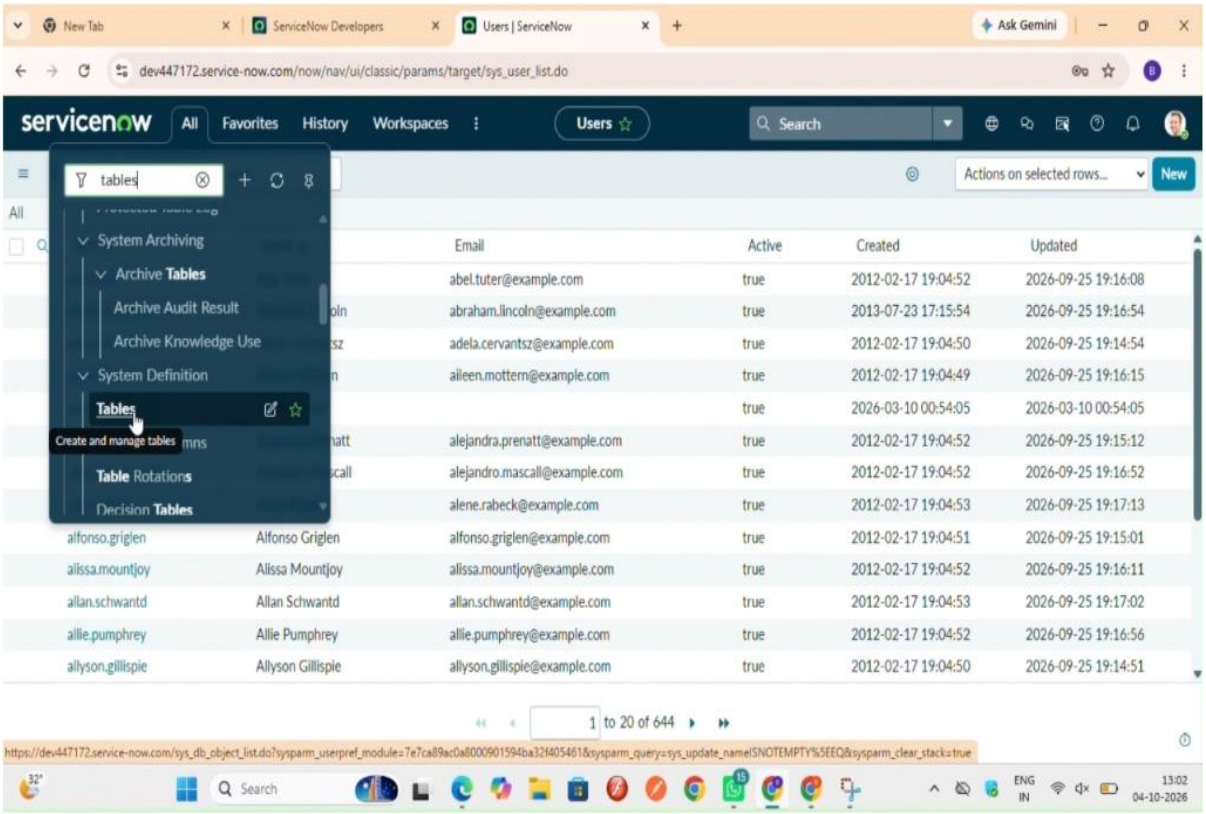
Creation of Table

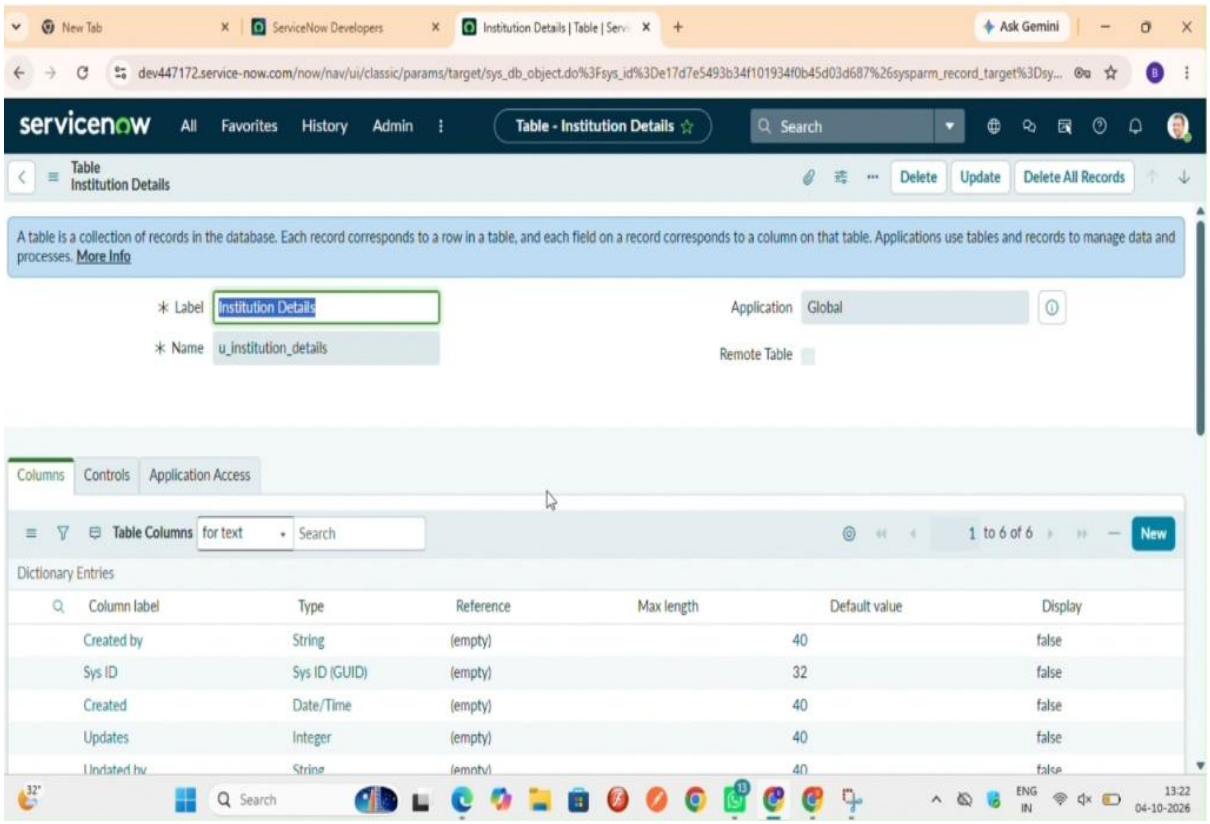
1. In the Application Navigator, search for System Definition → Tables.

2. Click New and create a custom table with the following details

   - **Label:** Institution Details
   - **Name:** u_institution_details
   - **Extends for :** false
   - Save or Submit the form

In the columns tab, user Insert a New row method to create the mentioned fields as shown below picture



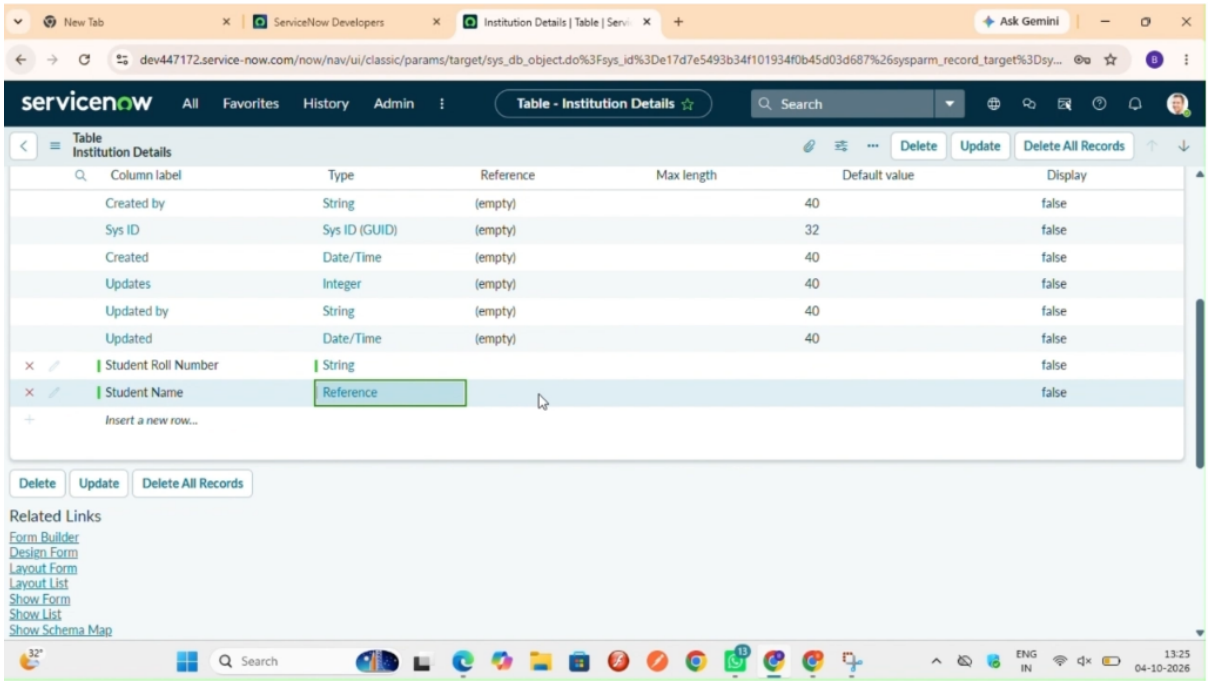


3.Click **Submit**

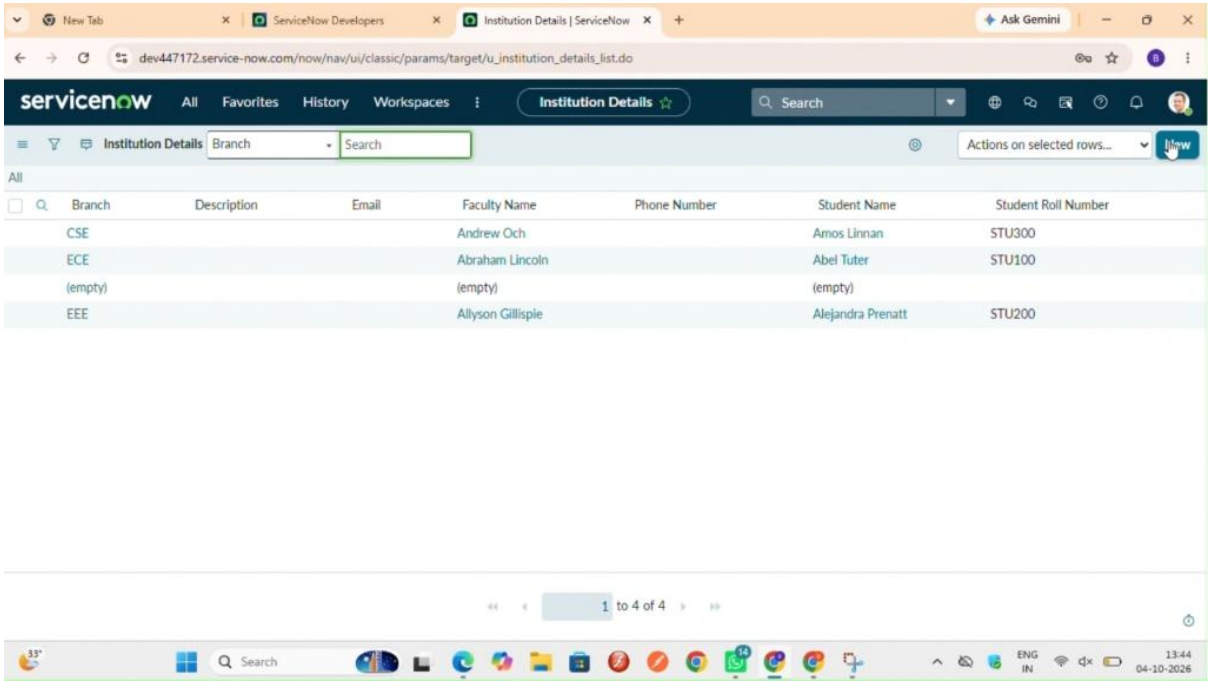
4. The following fields to the **u_institution_details** table

- **Student Roll Number**– Auto Number
- **Student Name** – Reference- User
- **Faculty Name** – Reference- User
- **Branch** – Choice (ECE, EEE, CSE)
- **Email**– String
- **Phone Number**– String
- **Description**– Multi String

5. Save the Table Click on the Show List in the Related Links



6. Create multiple records in the Student Records table with different branch values (ECE, EEE, CSE).



Milestone 3: Creation of Access Control List-READ

ACL Configuration

1. In the Application Navigator, search for System Security → Access Control (ACL).

2. Elevate “Security_admin” role

3. Click New to create a new ACL.

4. Enter the following details

   - **Type:** record
   - **Operation:** read
   - **Name:** u_institution_details
   - **Active:** true
   - **Advanced:** true

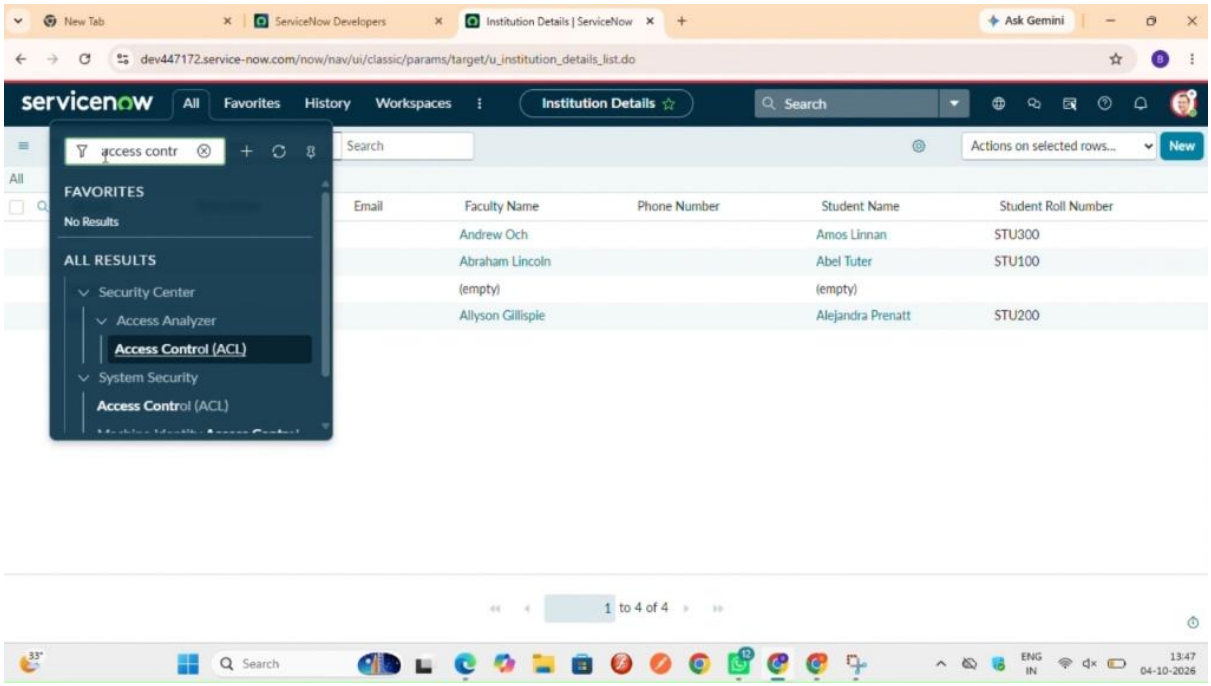
5. In the Requires role related list, add:

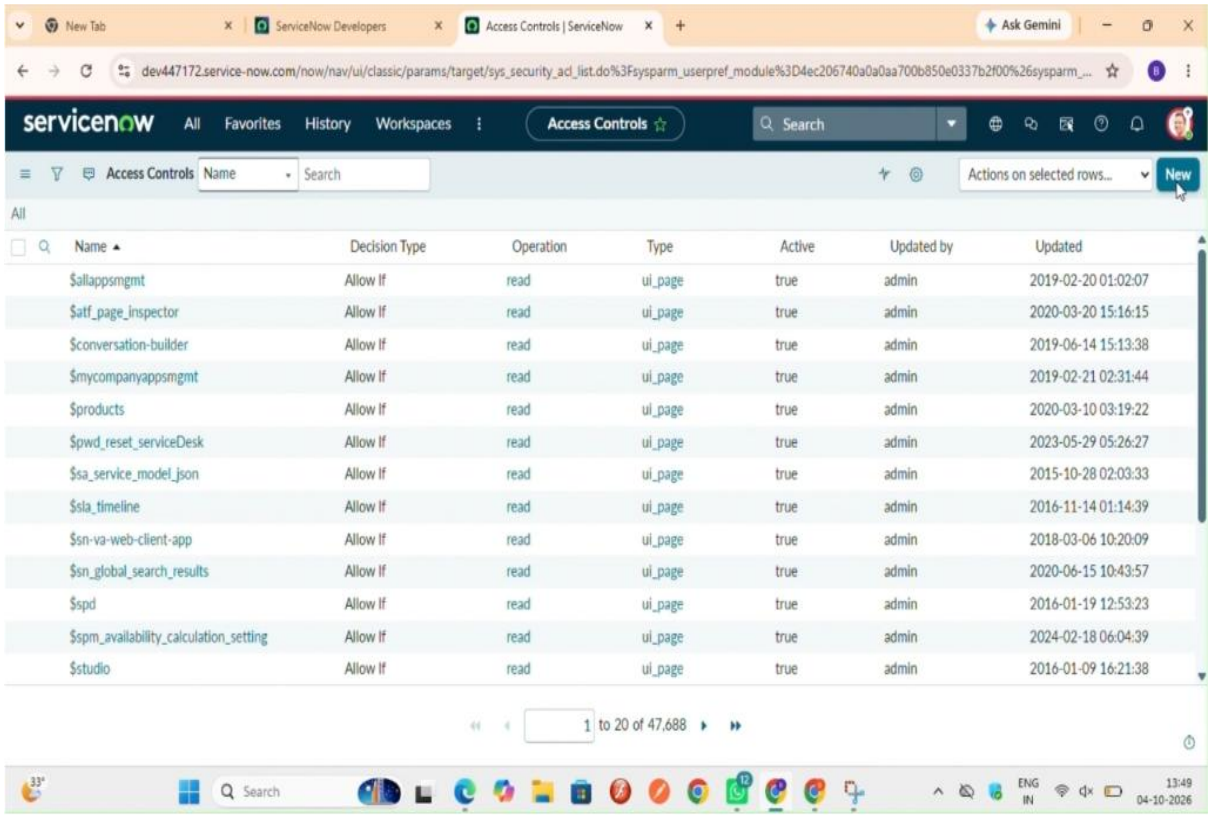
   - bb1 (custom role)

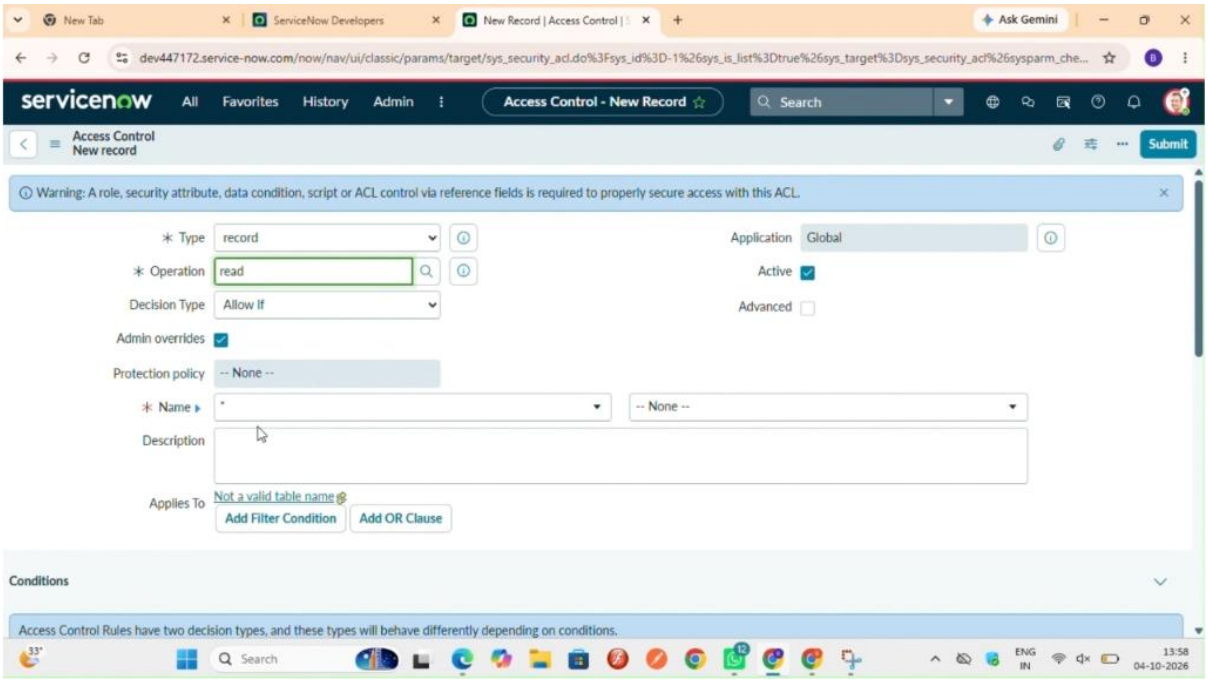
6. Add Data condition as Branch is EEE

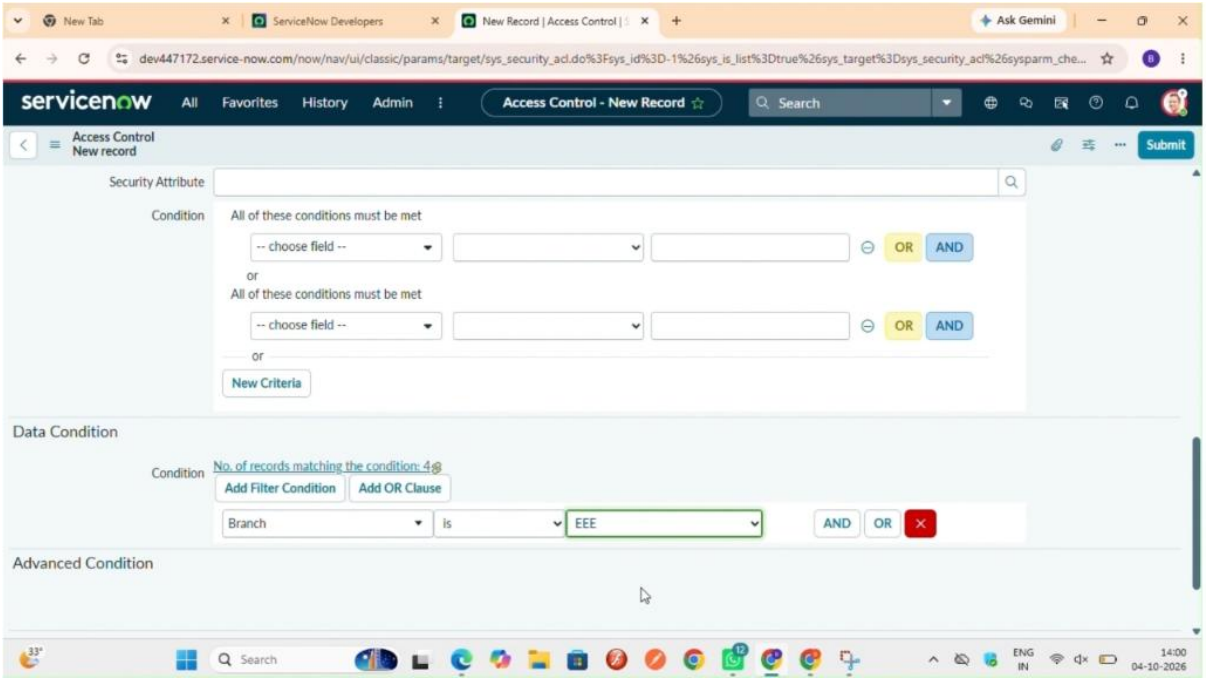
7. In the Script field, add the following script

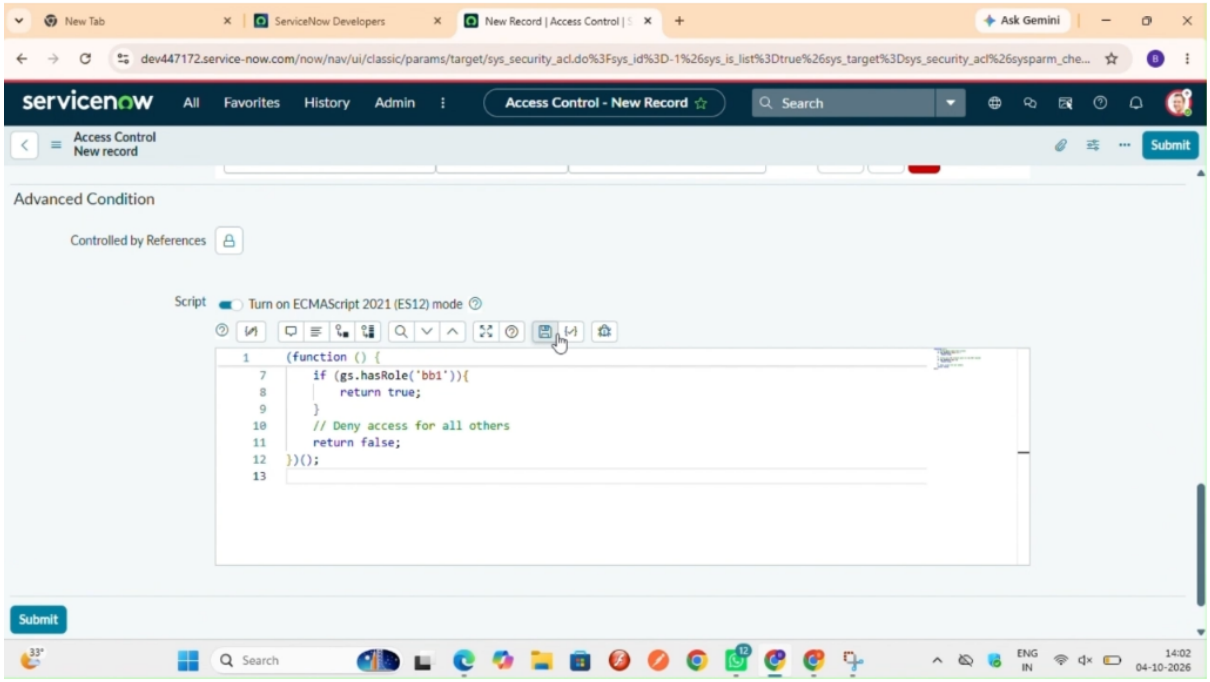
8. Save the ACL form


New Tab
ServiceNow Developers
Institution Details | ServiceNow
Ask Gemini
dev447172.service-now.com/now/nav/ui/classic/params/target/u_institution_details_list.do
servicenow
All
Favorites
History
Workspaces
Institution Details
Search
access contr
FAVORITES
No Results
ALL RESULTS
Security Center
Access Analyzer
Access Control (ACL)
System Security
Access Control (ACL)
Actions on selected rows...
New
Email
Faculty Name
Phone Number
Student Name
Student Roll Number
Andrew Och
Amos Linnan
STU300
Abraham Lincoln
Abel Tuter
STU100
(empty)
(empty)
Allyson Gillispie
Alejandra Prenatt
STU200
1 to 4 of 4
Search
ENG
IN
13:47
04-10-2026


New Tab
ServiceNow Developers
Access Controls | ServiceNow
Ask Gemini
dev447172.service-now.com/now/nav/ui/classic/params/target/sys_security_acl_list.do%3Fsysparm_userpref_module%3D4ec206740a0a0aa700b850e0337b2f00%26sysparm_...
servicenow
All
Favorites
History
Workspaces
Access Controls
Search
Access Controls
Name
Search
Actions on selected rows...
New
All
Name
Decision Type
Operation
Type
Active
Updated by
Updated
$allappsmgmt
Allow If
read
ui_page
true
admin
2019-02-20 01:02:07
$atf_page_inspector
Allow If
read
ui_page
true
admin
2020-03-20 15:16:15
$conversation-builder
Allow If
read
ui_page
true
admin
2019-06-14 15:13:38
$mycompanyappsmgmt
Allow If
read
ui_page
true
admin
2019-02-21 02:31:44
$products
Allow If
read
ui_page
true
admin
2020-03-10 03:19:22
$pwd_reset_serviceDesk
Allow If
read
ui_page
true
admin
2023-05-29 05:26:27
$sa_service_model_json
Allow If
read
ui_page
true
admin
2015-10-28 02:03:33
$sla_timeline
Allow If
read
ui_page
true
admin
2016-11-14 01:14:39
$sn-va-web-client-app
Allow If
read
ui_page
true
admin
2018-03-06 10:20:09
$sn_global_search_results
Allow If
read
ui_page
true
admin
2020-06-15 10:43:57
$spd
Allow If
read
ui_page
true
admin
2016-01-19 12:53:23
$spm_availability_calculation_setting
Allow If
read
ui_page
true
admin
2024-02-18 06:04:39
$studio
Allow If
read
ui_page
true
admin
2016-01-09 16:21:38
1 to 20 of 47,688
Search
ENG
IN
13:49
04-10-2026


New Tab
ServiceNow Developers
New Record | Access Control |
Ask Gemini
dev447172.service-now.com/now/nav/ui/classic/params/target/sys_security_acl.do%3Fsys_id%3D-1%26sys_is_list%3Dtrue%26sys_target%3Dsys_security_acl%26sysparm_che...
servicenow
All
Favorites
History
Admin
Access Control - New Record
Search
Access Control
New record
Submit
Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.
Type
record
Application
Global
Operation
read
Active
Decision Type
Allow If
Advanced
Admin overrides
Protection policy
-- None --
Name
-- None --
Description
Applies To
Not a valid table name
Add Filter Condition
Add OR Clause
Conditions
Access Control Rules have two decision types, and these types will behave differently depending on conditions.
Search
ENG
IN
13:58
04-10-2026


New Tab
ServiceNow Developers
New Record | Access Control |
Ask Gemini
dev447172.service-now.com/now/nav/ui/classic/params/target/sys_security_acl.do%3Fsys_id%3D-1%26sys_is_list%3Dtrue%26sys_target%3Dsys_security_acl%26sysparm_che...
servicenow
All
Favorites
History
Admin
Access Control - New Record
Search
Access Control
New record
Submit
Security Attribute
Condition
All of these conditions must be met
-- choose field --
OR
AND
or
All of these conditions must be met
-- choose field --
OR
AND
or
New Criteria
Data Condition
Condition
No. of records matching the condition: 4
Add Filter Condition
Add OR Clause
Branch
is
EEE
AND
OR
Advanced Condition
Search
ENG
IN
14:00
04-10-2026



9. Impersonate EEE users and open the Student Records list.

10. Copy the table name u_institution_details.list/LIST and paste in the All menu.

11. Impersonate Admin and open the Student Records list.

12. **Verify :** User with bb1 role can view only EEE branch records

13. User without the role cannot view any records

14. Admin user can view all records regardless of branch.

Milestone 4: Creation of Access Control List-CREATE

Creation of ACL-Create

1. Click New to create a new ACL(Create)

2. Enter the following details

3. Type: record

4. Operation: create

5. Name: u_institution_details

6. Active: true

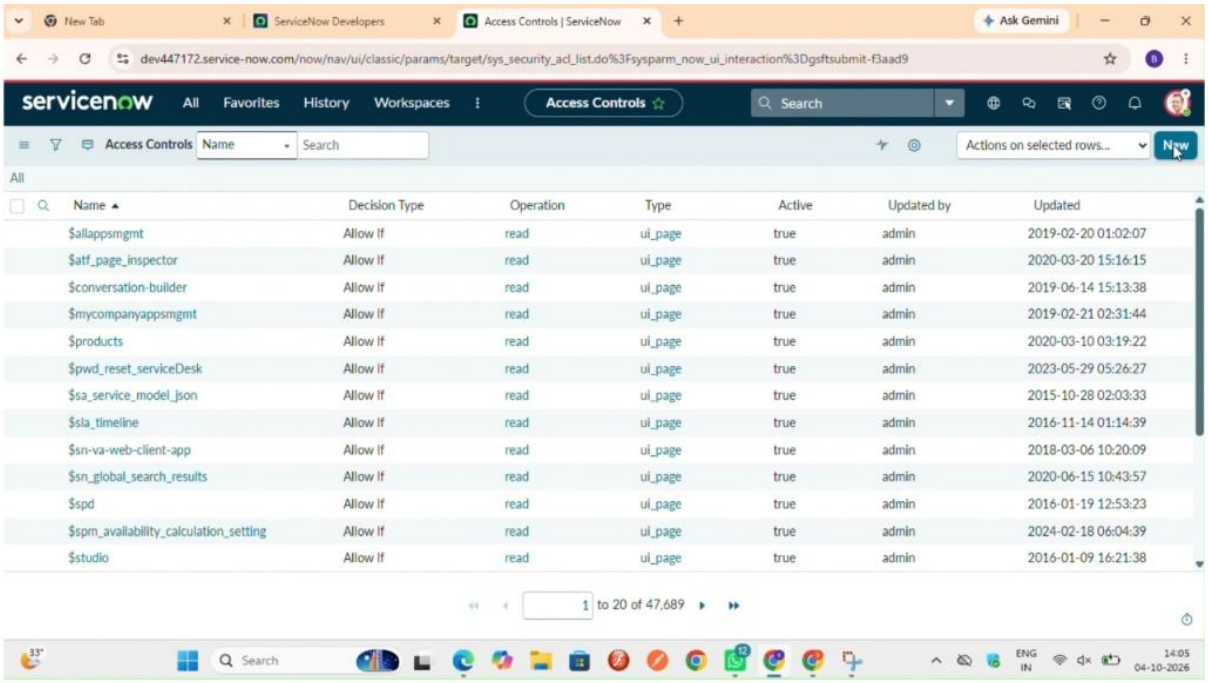
7. In the Requires role related list, add: bb2 (custom role)

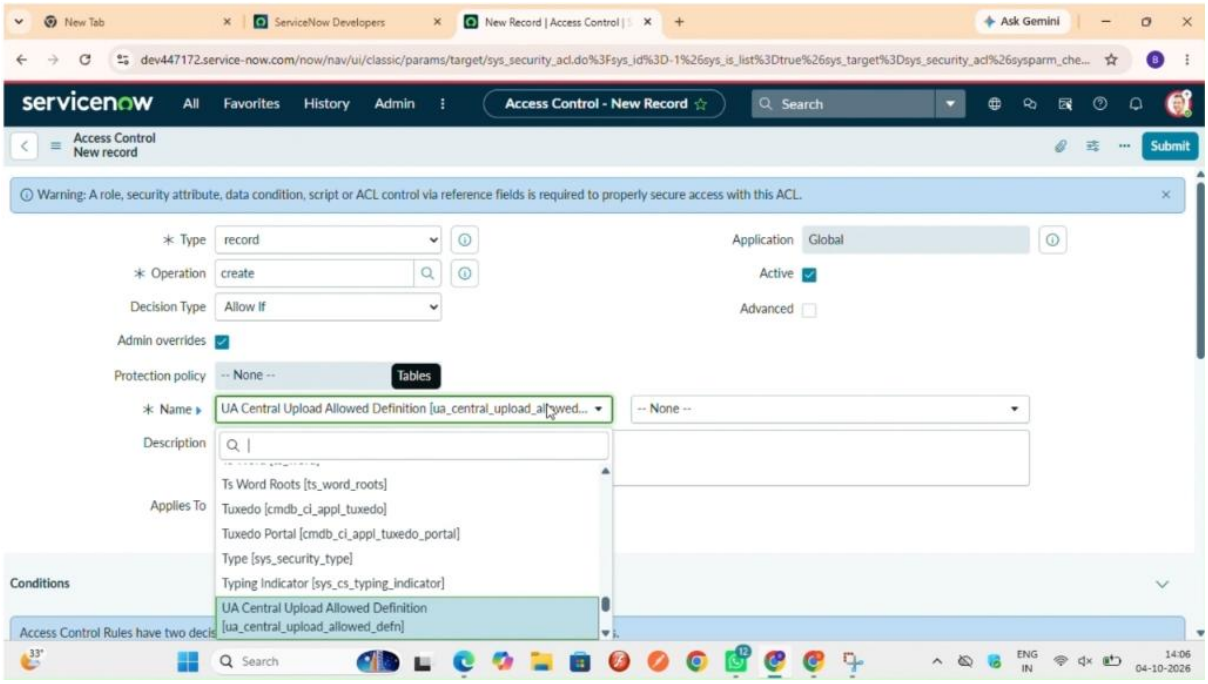
8. No need to add any Data condition

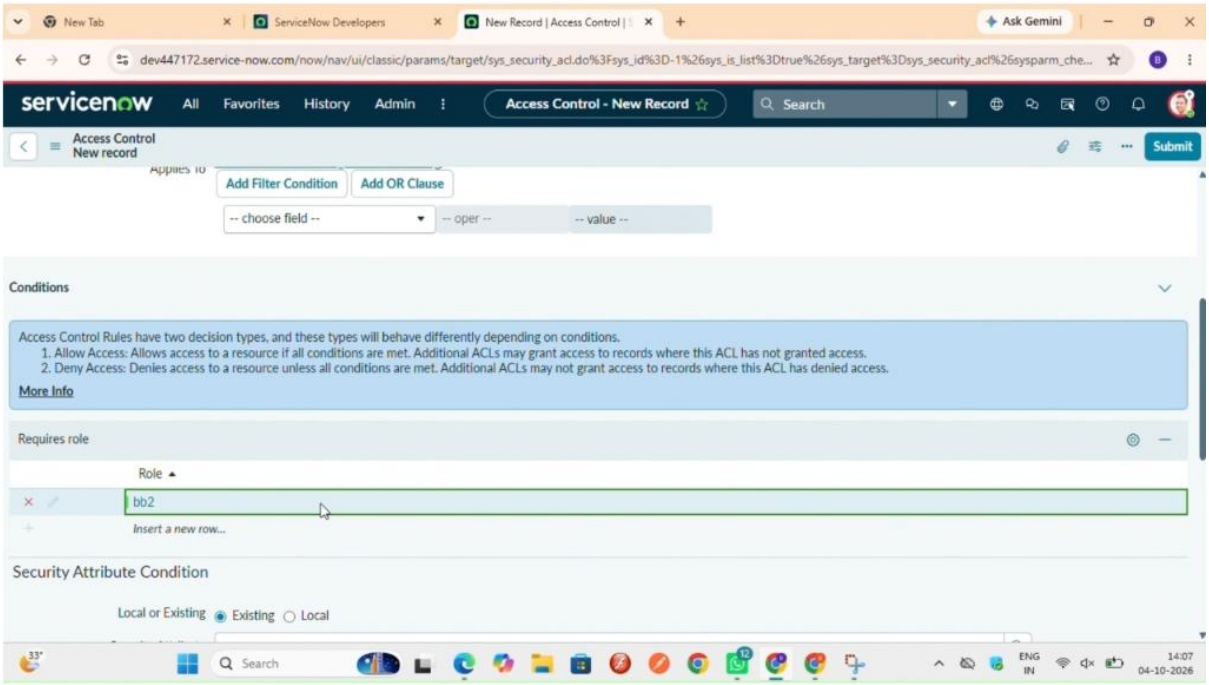
9. Save the ACL Record

10. Verify :

11. User with bb1 &bb2 roles can view only EEE branch records & New Button.







Milestone5: Creation of Access Control List-WRITE

Creation of ACL-Write

1. Click New to create a new ACL(Write)

2. Enter the following details

3. Type: record

4. Operation: Write

5. Name: u_institution_details

6. Active: true

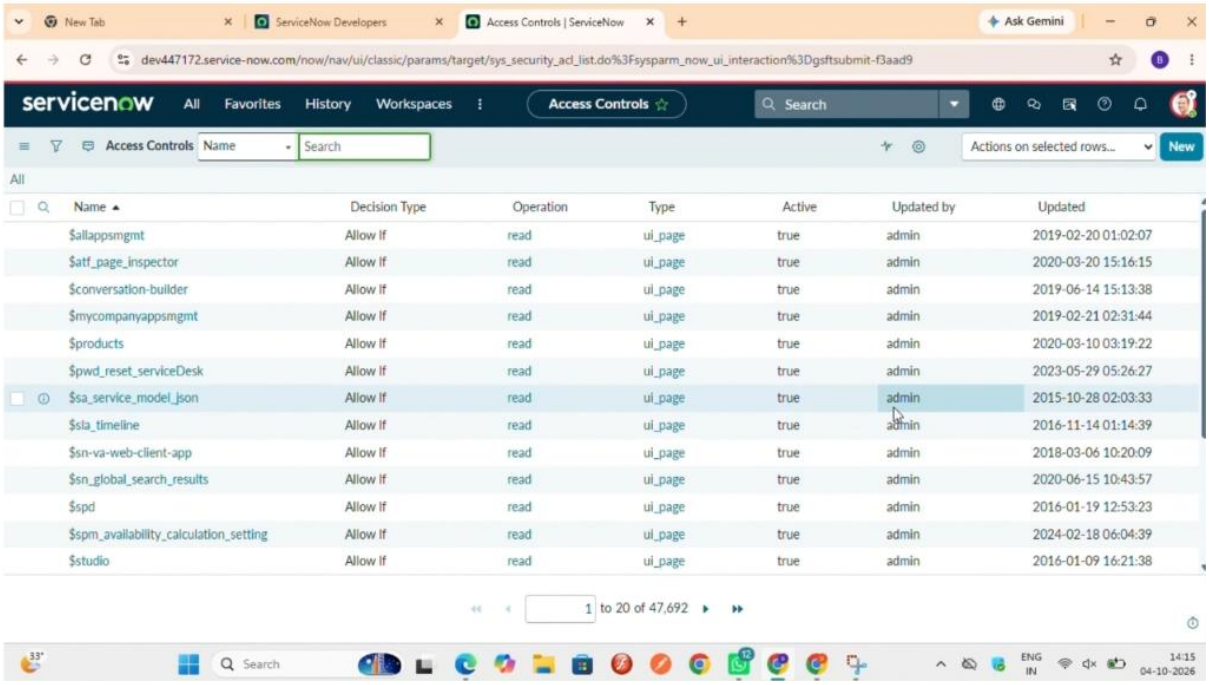
7. In the Requires role related list, add: bb3 (custom role)

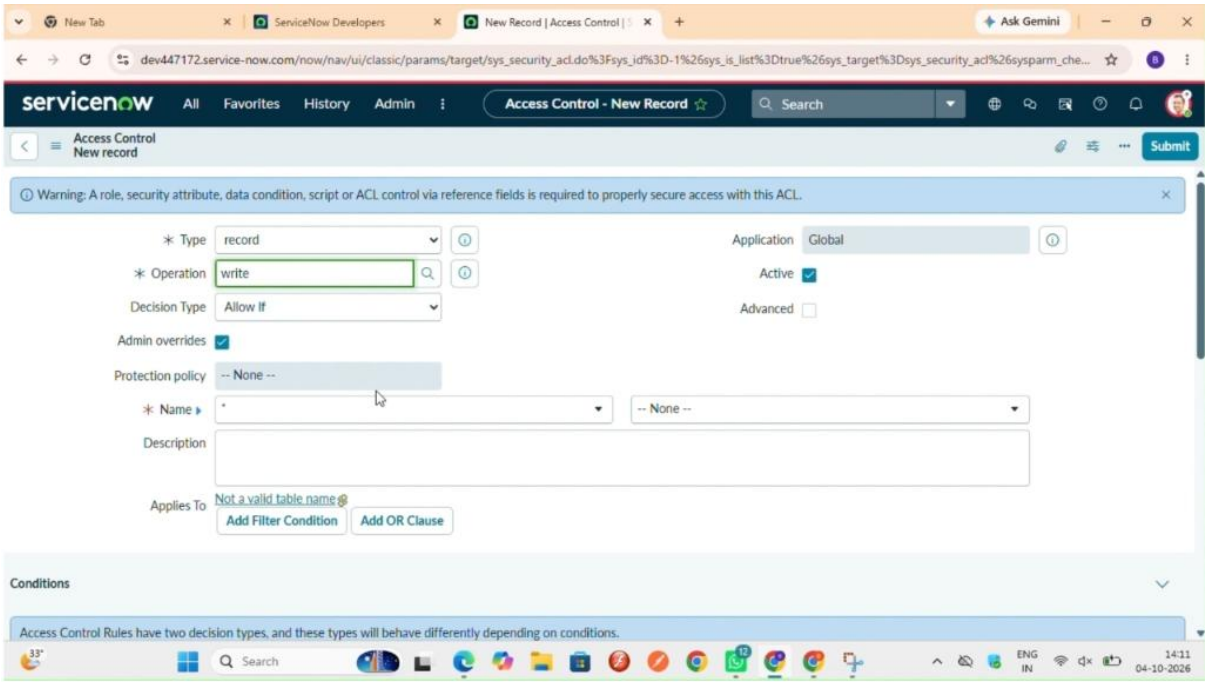
8. No need to add any Data condition

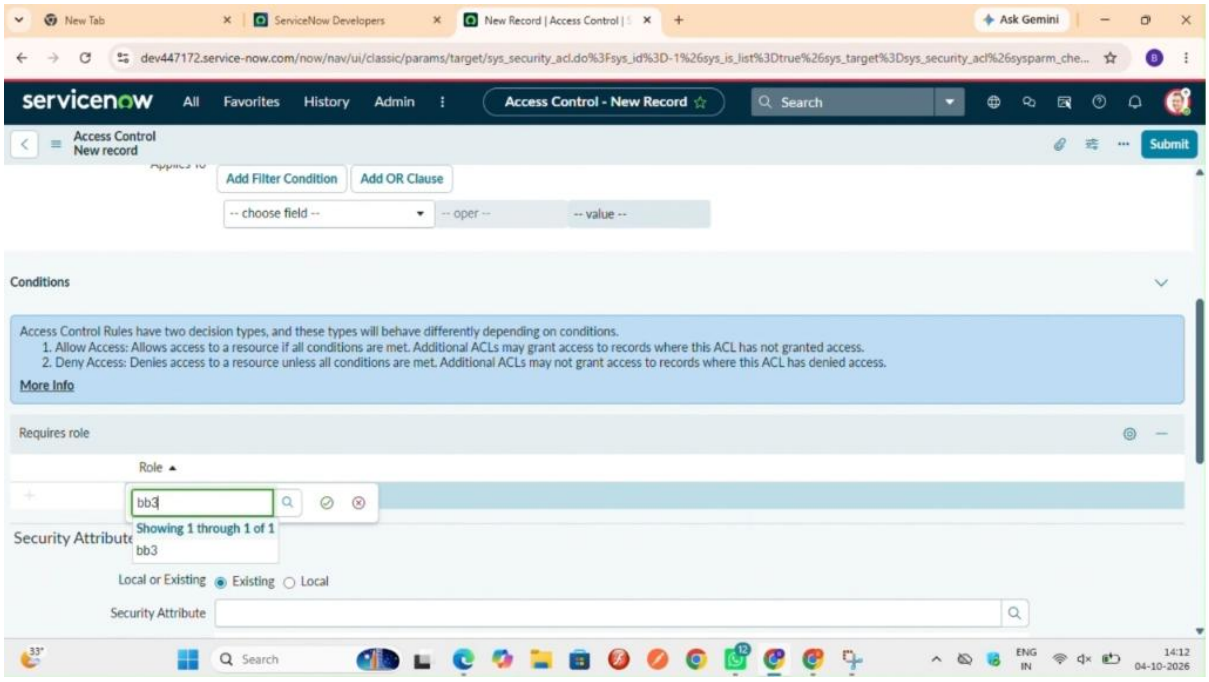
9. Save the ACL Record

10. Verify :

11. User with bb1 &bb2 &bb3 roles can view and edit the EEE branch records along with New Button.







Milestone6: Creation of Access Control List-DELETE

Creation of ACL-DELETE

1. Click New to create a new ACL(Delete)

2. Enter the following details

3. Type: record

4. Operation: Delete

5. Name: u_institution_details

6. Active: true

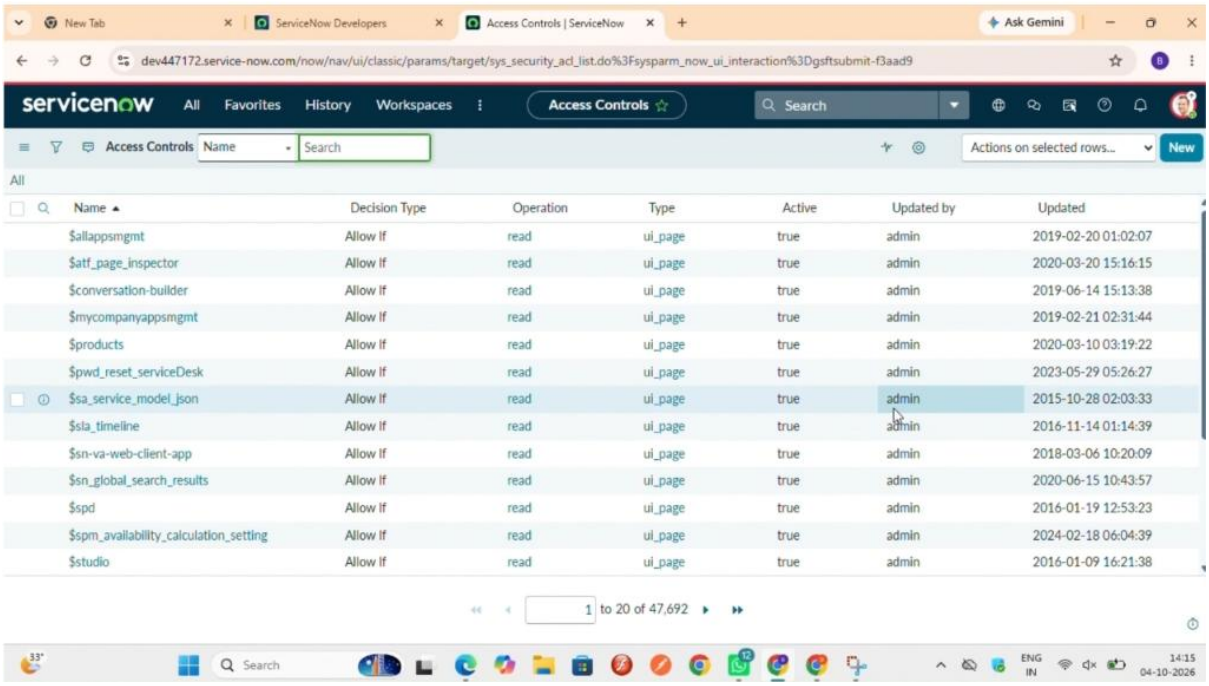
7. In the Requires role related list, add: bb4 (custom role)

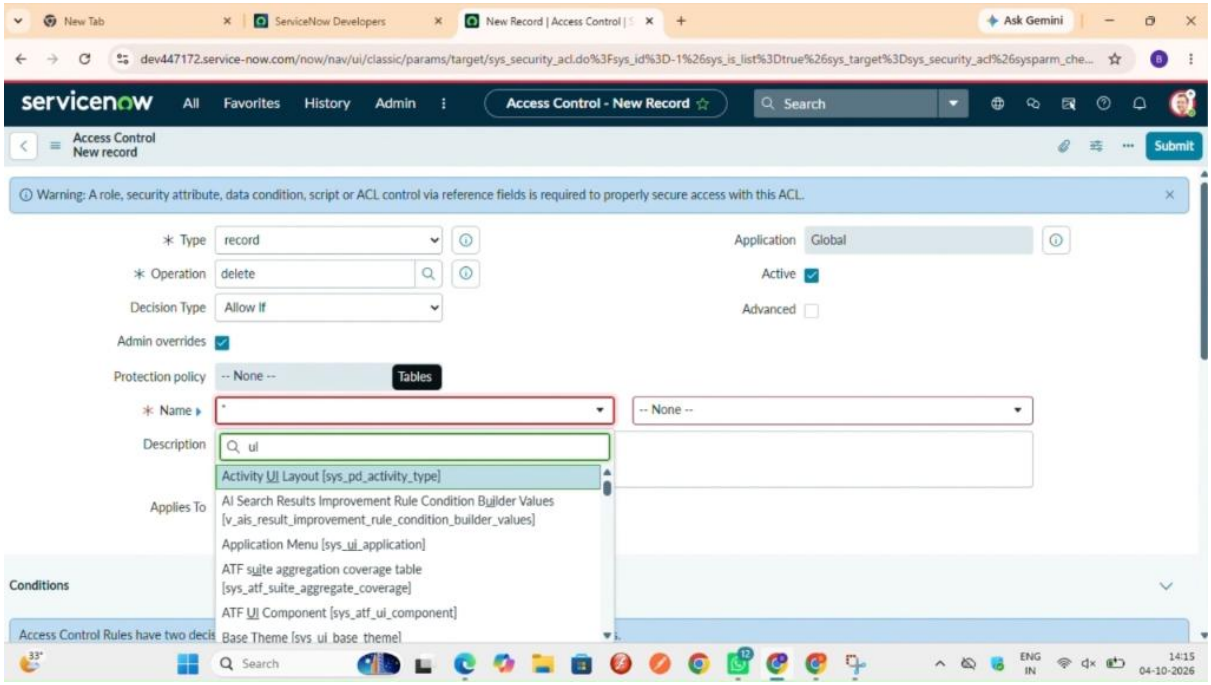
8. No need to add any Data condition

9. Save the ACL Record

10. Verify :

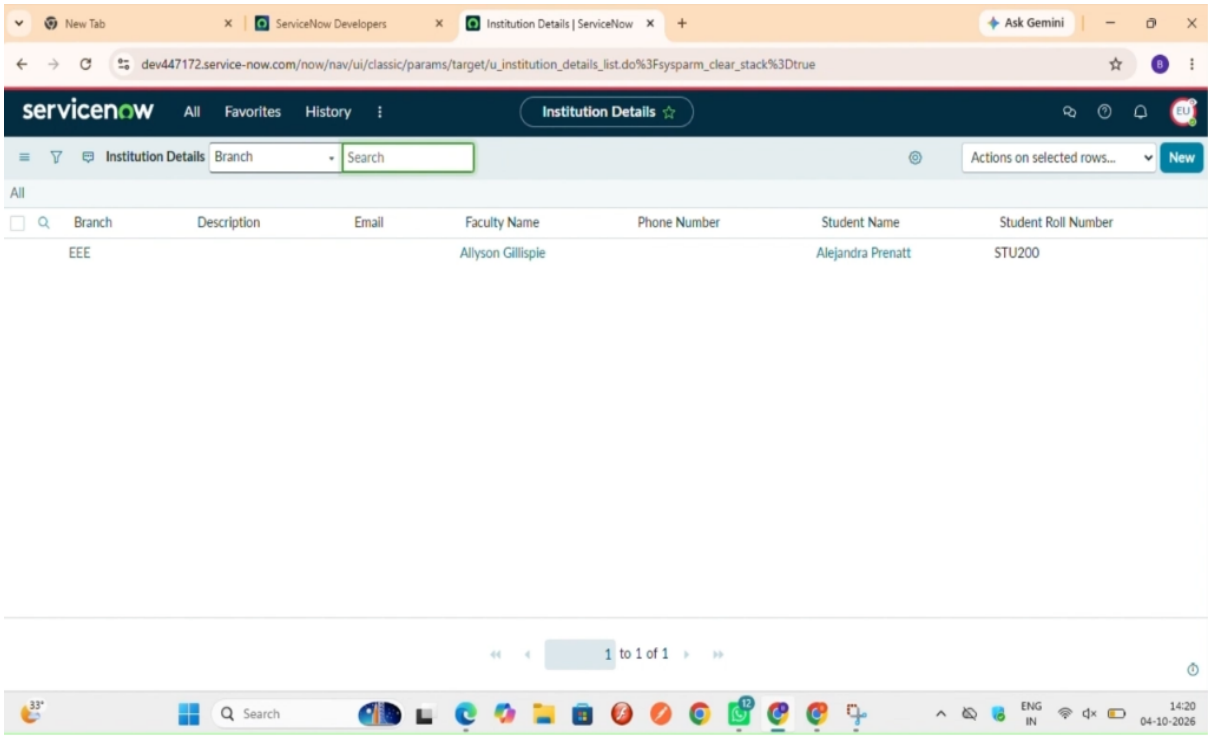
11. Users with bb1 &bb2 &bb3&bb4 roles can see, create and edit along with delete the EEE branch records.





Milestone7: Conclusion

Project Outcome:



Description

By completing this lab, learners gain a comprehensive understanding of how script-controlled READ, WRITE, CREATE, and DELETE ACLs enforce record-level security, how user roles and record field values are evaluated together to make access decisions, and how ACLs protect sensitive data from unauthorized viewing, modification, creation, and deletion across forms, lists, and Service Portal interfaces.